

MATTHEW MURRAY

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Education

University of Waterloo

Sep. 2025 – May 2030

Candidate for a Bachelor of Applied Sciences of Biomedical Engineering

Experience

University of Calgary, Department of Clinical Neurosciences and Radiology

May 2026 – August 2026

Research Assistant

- Designed and conducted 20+ qualitative interviews to identify the informational needs of patients with degenerative cervical myelopathy.
- Analyzed qualitative interview data using thematic coding to guide the design of a personalized clinical reporting system.
- Designed a patient-facing precision medicine report in Figma to improve communication of individualized clinical data.

Leadership & Community Impact

Engineering Club

Sept 2023 – June 2025

Co-President

- Directed weekly meetings and programming for an 80+ member engineering club, tripling active membership over two years.
- Managed club budget, communications, and event logistics using Excel, Microsoft Office, and Google Workspace.
- Supervised the executive team by delegating responsibilities and coordinating weekly operations.

Head Buddies

March 2024 – June 2025

Co-Lead

- Co-designed the application and interview process, recruiting and training a leadership team of 70+ senior students.
- Led 12 team leaders in delivering weekly classroom sessions on leadership, equity, bullying, peer pressure, and study habits.
- Planned and coordinated programming that supported the transition of more than 250 Grade 9 students into high school.

Projects

Accessibility Devices | SolidWorks, Bambu Studio, 3D Printing

December 2025 – February 2026

- Designed and manufactured a custom key holder in SolidWorks to improve usability for individuals with arthritis.
- Iteratively prototyped and evaluated grip force, hand placement, and then manufacturability using Bambu Studio.
- Created and manufactured Braille keyboard keycaps to support an individual with visual impairment.

Board Game Design for Cerebral Palsy | QFD, Pairwise, BOC, SWOT

September 2025 – December 2025

- Applied a user-centered engineering design process to increase board game accessibility for children aged 8-11 with spastic cerebral palsy.
- Applied QFD, Pairwise, BOC, and SWOT analyses to translate user needs into design requirements.
- Developed and justified a design solution tailored to the physical and cognitive needs of the target user group.

Skills

Design & CAD: SolidWorks, Z-Suite, Bambu Studio, Figma, Canva, Human-centered design, QFD, Pairwise Analysis, BOC Analysis, SWOT Analysis

Programming: C++, MATLAB, HTML & JavaScript & CSS (currently learning)

Software: Excel (PivotTables, data analysis, Generative AI), PowerPoint, Word, Google Workspace, VS Code, Clion

Lab & Research: Literature reviews, scoping reviews, micropipetting, academic writing, clinical interviews

Certifications: National Lifeguard, Standard First Aid & CPR-C, WHMIS, TCPS 2

Interpersonal: Leadership, public speaking, interdisciplinary collaboration, project coordination, ENG SOC rep, Marathon and Ironman 70.3 finisher, fluent in English and French

Awards

- **David Phillips Carpe Diem Award** – Outstanding leadership, community service, and positive community impact.
- **Leamen Memorial Award** – Outstanding citizenship.
- **Valedictorian, Shad UPEI '24**